

ORGANIZATIONAL MATURITY ASSESSMENT (OMA) – BACKEND ARCHITECTURE MAP

Note

This is a high-level backend system map.

OMA is a **diagnostic intelligence engine** feeding HR Strategy, not a standalone report.

Final scope, timelines, and cost depend on phase prioritization.

1. CORE SERVICES

A. API Gateway

- Request routing
 - Authentication & Authorization
 - Rate limiting
 - Request logging
 - CORS handling
 - Payload validation
 - Survey anonymity enforcement middleware
 - Benchmark data access control
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B. Authentication Service

- User login / logout
 - JWT token generation
 - Role-based access control (HR, Executive, Manager, Employee, Auditor)
 - Session management
 - Survey participation tokens
 - Read vs Analyze permission enforcement
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2. BUSINESS LOGIC SERVICES

A. Survey Design & Launch Service

Endpoints

POST /api/oma/surveys

POST /api/oma/surveys/{id}/launch

GET /api/oma/surveys/{id}

GET /api/oma/surveys/{id}/status

Business Logic

- Separate surveys per respondent group (HR / Managers / Employees)
 - Surveys mapped to predefined maturity domains
 - Anonymous participation support
 - Configurable launch window
 - Survey lifecycle tracking
 - Audit logging of survey launch
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B. Survey Response Collection Service

Endpoints

POST /api/oma/responses

GET /api/oma/responses/{surveyId}

Business Logic

- Secure response ingestion
 - Responses stored by respondent group
 - Partial completion supported
 - Submission timestamps captured
 - Responses anonymized by default
 - Encryption at rest & in transit
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C. Data Normalization Service

Endpoints

POST /api/oma/normalize/{surveyId}

GET /api/oma/normalized/{surveyId}

Business Logic

- Normalize scale-based answers to standard range
 - Prepare free-text responses for AI processing
 - Detect & flag outliers
 - Normalized dataset versioned
 - Raw data preserved for audit
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D. Domain Scoring Engine

Endpoints

POST /api/oma/score/domains

GET /api/oma/scores/domains

Business Logic

- Domain-level maturity scoring (1–5)
 - Weighted scoring rules
 - Domain weights configurable per organization
 - Timestamped score storage
 - Historical comparison enabled
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E. Consistency & Validation Rules Engine

Endpoints

POST /api/oma/validate

GET /api/oma/validation/results

Business Logic

- Detect contradictions between respondent groups
 - Identify abnormal variance patterns
 - Apply predefined validation rules
 - Store validation results alongside scores
 - Flag governance & culture risks
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F. Overall Maturity Level Service

Endpoints

POST /api/oma/maturity/calculate

GET /api/oma/maturity

Business Logic

- Aggregate domain scores
- Assign maturity level:
 - Level 1 – Initial
 - Level 2 – Emerging
 - Level 3 – Defined

- Level 4 – Managed
 - Level 5 – Optimized
 - Calculation logic configurable
 - Final maturity level lock after confirmation
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G. Gap & Risk Identification Service

Endpoints

GET /api/oma/gaps

GET /api/oma/risks

Business Logic

- Identify gaps per domain
 - Assign risk severity (Low / Medium / High)
 - Generate gap explanations
 - Link gaps to maturity drivers
 - Store actionable diagnostic insights
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H. Improvement Roadmap Generator

Endpoints

POST /api/oma/roadmap/generate

PUT /api/oma/roadmap/{id}

GET /api/oma/roadmap

Business Logic

- Generate improvement actions per gap
 - Prioritize by impact vs effort
 - Roadmap structured quarterly (6–12 months)
 - Editable by HR
 - Roadmap version-controlled
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I. Benchmarking Engine (Industry / Size / Region)

Endpoints

POST /api/oma/benchmark/classify

POST /api/oma/benchmark/apply
GET /api/oma/benchmark/report

Business Logic

- Company classification:
 - Industry
 - Size
 - Region
 - Apply benchmark datasets
 - Domain-level comparison
 - Variance calculation (Above / At / Below)
 - Benchmark dataset version tracking
 - Enforce benchmark anonymization
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J. Strategy Integration Service

Endpoints

POST /api/oma/publish/strategy
GET /api/oma/strategy/ack

Business Logic

- Send maturity results to HR Strategy Engine:
 - Overall maturity level
 - Domain scores
 - Gaps & risks
 - Transmission acknowledgment
 - Integration event logging
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3. INTEGRATION SERVICES

A. AI Processing Service

- Free-text response analysis
- Gap explanation generation
- Roadmap action synthesis

B. HR Strategy Engine Integration

- Readiness input
- Gap-driven strategy alignment
- Feedback loop validation

C. Notification Service

- Survey invitations
- Survey reminders
- Results readiness notifications

D. Reporting & Export Service

- Benchmark-enhanced maturity reports
- Domain heatmaps
- PDF / PPT export

4. DATABASE (SUMMARY)

- OMA_Surveys
 - OMA_Survey_Questions
 - OMA_Survey_Responses
 - OMA_Normalized_Data
 - OMA_Domain_Scores
 - OMA_Validation_Results
 - OMA_Maturity_Levels
 - OMA_Gaps
 - OMA_Risks
 - OMA_Roadmaps
 - OMA_Benchmarks
 - OMA_Benchmark_Datasets
 - OMA_Strategy_Exports
 - Audit_Logs
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DATABASE ARCHITECTURE

A. OMA_Surveys

- id (PK)
 - respondent_group (HR / Manager / Employee)
 - launch_start
 - launch_end
 - status
 - created_at
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B. OMA_Survey_Responses

- id (PK)
 - survey_id (FK)
 - response_payload (JSON)
 - submitted_at
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C. OMA_Domain_Scores

- id (PK)
 - domain_name
 - score
 - calculated_at
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D. OMA_Maturity_Levels

- id (PK)
 - overall_level
 - confirmed_at
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E. OMA_Gaps

- id (PK)
- domain_name
- gap_description
- severity

F. OMA_Roadmaps

- id (PK)
- quarter
- actions (JSON)
- version

G. OMA_Benchmarks

- id (PK)
- benchmark_type (Industry / Size / Region)
- variance_data (JSON)
- dataset_version

USER STORY COVERAGE

- ☒ EPIC 1 – Organizational Maturity Assessment
- ☒ EPIC 2 – Benchmarking Integration
- ☒ Diagnostic → Actionable Roadmap
- ☒ Strategy Feedback Loop